

CS109 Midterm Examination

This is a closed calculator/computer exam. You are, however, allowed to use notes in the exam. The last page of the exam is a Standard Normal Table, in case you need it.

You have 2 hours (120 minutes) to take the exam. The exam is 120 points, meant to roughly correspond to one point per minute of the exam. You may want to use the point allocation for each problem as an indicator for pacing yourself on the exam.

In the event of an incorrect answer, any explanation you provide of how you obtained your answer can potentially allow us to give you partial credit for a problem. For example, describe the distributions and parameter values you used, where appropriate. It is fine for your answers to include summations, products, factorials, exponentials, and combinations, unless the question specifically asks for a numeric quantity or closed form. Where numeric answers are required, the use of fractions is fine.

Problem	Score
1 (17 pts)	
2 (16 pts)	
3 (26 pts)	
4 (20 pts)	
5 (16 pts)	
6 (25 pts)	
Total (120 pts)	

THE STANFORD UNIVERSITY HONOR CODE

- A. The Honor Code is an undertaking of the students, individually and collectively:
- (1) that they will not give or receive aid in examinations; that they will not give or receive unpermitted aid in class work, in the preparation of reports, or in any other work that is to be used by the instructor as the basis of grading;
 - (2) that they will do their share and take an active part in seeing to it that others as well as themselves uphold the spirit and letter of the Honor Code.
- B. The faculty on its part manifests its confidence in the honor of its students by refraining from proctoring examinations and from taking unusual and unreasonable precautions to prevent the forms of dishonesty mentioned above. The faculty will also avoid as far as practicable, academic procedures that create temptations to violate the Honor Code.
- C. While the faculty alone has the right and obligation to set academic requirements, the students and faculty will work together to create optimal conditions for honorable academic work.

I acknowledge and accept the letter and spirit of the honor code:

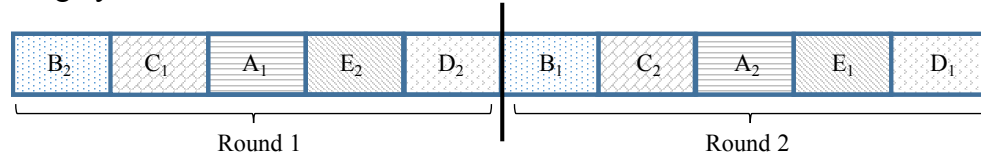
Signature: _____

Name (print): _____

- c. (5 points) What is the probability that a random ordering of the songs has two songs from artist A in a row or two songs from artist B in a row?

- d. (4 points) Spotify tries an algorithm where they play the 10 songs in two rounds:
- In the first round, they play one song from each artist.
 - In the second round, they play the other song from each artist, preserving the **same artist ordering** as in the first round.

Here is an example of one ordering produced by the algorithm. A_2 is the second song by artist A:



How many ways are there to order songs under this new algorithm?

Spotify uses a shuffling algorithm like the one in part (d) for the reason described in (b)

- c) (8 points) To slow down the hacker you make your website wait 2^n seconds before responding to a login request, where n is the number of times a visitor has incorrectly guessed their password. If the hacker has a $1/1000$ chance of correctly guessing a password on each attempt, what is the expected amount of time it will take the hacker to guess a user's password?

The strategy used in part (d) is a great password protection algorithm.

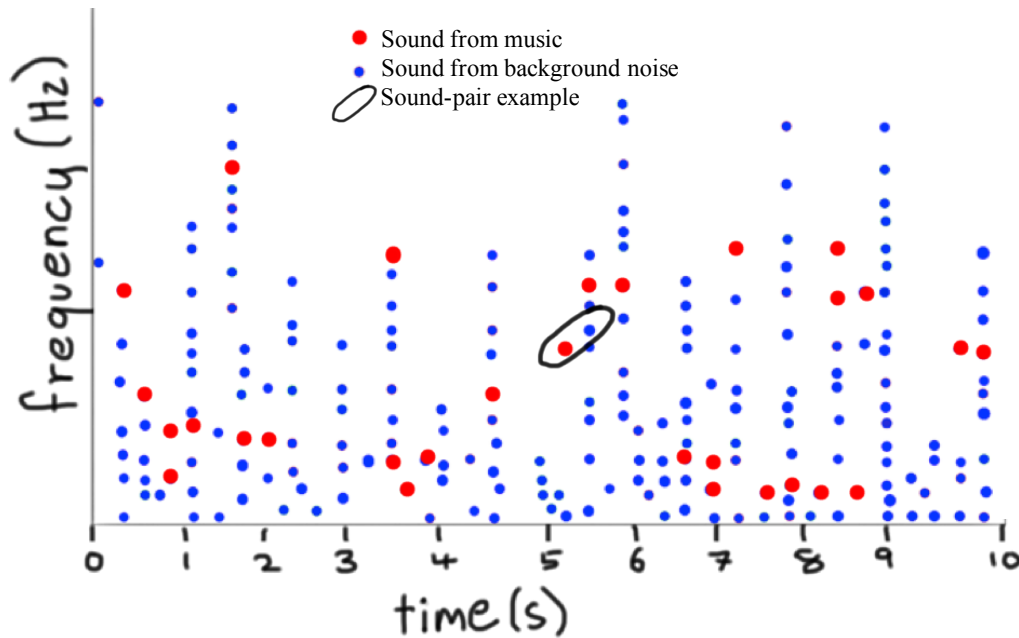
3. Shazam (26 points)

Shazam is an app that listens to a 10 second sample of background music playing (for example in a restaurant) and guesses the song. A user has just sent in a sample which we will use to explore how Shazam works. Within the 10 second sample there are:

50 sounds heard from the background **music** and

2000 sounds heard from background **noise**

A “sound” is a frequency heard at a particular time. Here is a visualization of our 10 second sample (a few sounds are omitted for visual clarity):



- (3 points) There are 2050 total sounds in the sample. How many distinct **pairs** of two sounds are there? Sounds can not be paired with themselves. Sound-pairs with the same two sounds are **not** distinct.
- (4 points) How many distinct sound pairs exist such that **both** sounds in the sound-pair are from the music (as opposed to being from background noise)?

- c. (5 points) Every distinct pair of sounds from the 10 second sample casts a vote as to what song the pair thinks is playing. If both sounds in a sound-pair are from the music, the pair always casts a vote for the correct song. Otherwise, since the pair contains background noise, the pair casts a vote uniformly at random from a set of 5 songs (always the same five songs, including the correct song).

We want **more than $1/5$ of the total number of votes** to go to the correct song. How many of the pairs containing background noise must vote for the correct song in order for the correct song to get $1/5$ of the votes?

- d. (6 points) Let d be the number of sound-pairs that you calculated in part (c). What is the probability that the correct song receives more than $1/5$ of all the votes?

- e. (8 points) You are at the Tree House and hear music playing. You believe that there is an:

80% chance the song is: Hold Up by Beyonce (event X_1)

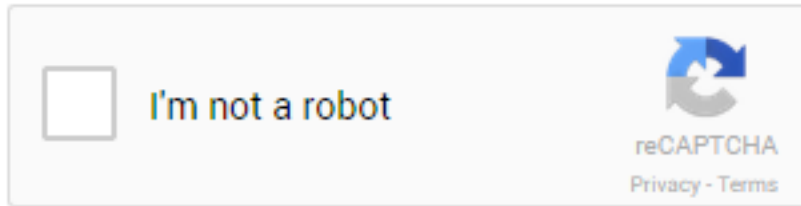
20% chance the song is: Can't Get Used by Andy Williams (event X_2)

You run "Shazam" and the app returns that it predicts the Andy Williams song is playing. Let q be the probability that the correct song is returned by Shazam. Let $(1 - q)$ be the probability that an incorrect song is returned. What is your new probability for X_1 and X_2 ? Express your answer in terms of q .

This is the basic version of Shazam. Wisdom of the crowds is a phenomenon where a crowd (in this case a crowd of note-pairs) is correct consistently and wrong randomly.

4 Recaptcha (20 points)

Based on browser history, Google believes that **there is a 0.2 probability that a particular visitor to a website is a robot**. They decide to give the visitor a recaptcha:



Google presents the visitor with a box, 10 pixels wide by 10 pixels tall. The visitor must click inside the box to show that they are not a robot.

Google has observed that robots click very close to the center of a recaptcha. The distance D of a robot click from the center of the box, in pixels, is normally distributed with mean 0 and variance 2. Humans, on the other hand, click uniformly in the box (all locations are equally likely).

- a. (6 points) What is the probability that a robot clicks on a pixel that has a **distance from the center** of the box which is greater than or equal to 1.2 pixels?

- b. (6 points) What the the Probability Density Function (PDF) of a human clicking X pixels from the left of the box and Y pixels from the top of the box?

- c. (8 points) The visitor clicks in the box at pixel (3,3) which has a distance of 2 pixels from the center of the box. What is Google's **new** belief that the visitor is a robot?

Recaptcha uses more sophisticated statistics of natural human mouse gestures and clicks, but this problem covers the central idea behind the new click based recaptchas.

5. Exponentials (16 points)

- a. (8 points) What is the probability that an exponential random variable $X \sim \text{Exp}(\lambda)$ takes on a value that is within one standard deviation of its mean?
- b. (8 points) Let $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$ be independent exponential random variables. Let $M = \max(X_1, X_2)$ where $\max$ is a function which returns the larger of the two values. Give an expression for the cumulative density function of M .

- b. (7 points) Write an expression for the **exact** number of machines that you need (K) such that the probability that you don't drop a single request in the busiest hour is > 0.99 . Use p from part a. The expression does *not* have to be closed form.

- c. (7 points) A Poisson random variable with a parameter of $\lambda \geq 1000$, can be approximated by a normal. The approximating normal should match the mean and the variance of the Poisson. Let $X \sim \text{Poi}(1000)$. What is an approximation for $P(990 < X < 1000)$. Give your answer to two decimal places.
- d. (6 points) The probit function, $\Phi^{-1}(x)$ is the inverse of the CDF of a standard normal. It maps from probabilities to the standard normal CDF input that would produce said probability. For example, you can confirm using the Standard Normal Table that $\Phi(0.1) = 0.5398$, so $\Phi^{-1}(0.5398) = 0.1$. Give a closed form expression, using the probit function, for an approximation of K.

Websites solve this problem constantly.

CS109 Midterm Solutions

1. (a) The number of ways to order 10 distinct objects is $10!$. Common mistake: treating songs by the same artist as indistinguishable.
- (b) Group the two songs by artist A. There are 2 ways to order this group. Then if this group is treated as a single object, we have to order 9 objects, which can be done in $9!$ ways. This gives a total of $2 \cdot 9!$ ways to order the songs such that the two songs from artist A are played in a row. Divide by the sample space (from part a) to get

$$\frac{2 \cdot 9!}{10!} = 1/5$$

Common mistakes: forgetting to count the 2 ways of ordering the two songs by artist A. Only counting the number of ways to place the songs by artist A and ignoring the ordering of the rest of the songs. Only computing the probability of the two songs playing first (instead of in a row anywhere among the 10 songs).

- (c) Let A be the event that the two songs by artist A are played in a row, and let B be the event that the two songs by artist B are played in a row. Using the inclusion-exclusion principle, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$P(A) = P(B) = 1/5$ from part (b). To compute $P(A \cap B)$, we follow a similar approach to part (b): First, group the two songs by A in one group and the two songs by B in another group. For each of these two groups, there are 2 ways to order the two songs. Then this leaves us with 8 distinct things to order (the 6 other songs, the group of the two A songs, and the group of the two B songs), which can be done $8!$ ways. So $P(A \cap B) = 2 \cdot 2 \cdot 8!/10!$, and

$$P(A \cup B) = 1/5 + 1/5 - 2 \cdot 2 \cdot 8!/10! = 2/5 - 2/45 = 16/45$$

Common mistakes: just multiplying the answer to part (b) by 2. This only works if A and B are mutually exclusive, which they are not. Claiming $P(A \cap B) = P(A) \cdot P(B)$, which is not true here because A and B are not independent. Many people were slightly off in their computations of $P(A \cap B)$, miscounting the final number of things to order, or forgetting to order the 2-song groups, etc.

- (d) There are $5!$ ways to arrange the 5 artists in the first 5 songs. This determines the order of artists in the last five songs, so there is only one way to choose that order. For each artist, we must order the artist's two songs to decide which song to put in the first half ($2! = 2$ choices). Therefore, the total number of arrangements is $5! \cdot (2!)^5$.

Common mistake: leaving out the ordering of the songs, or only multiplying by $2!$ once.

2. (a) 10^8 . There are 10 options for each of the 8 digits in the passcode.
- (b) $\frac{10^6}{10^8} = 0.01$. They can try 10^6 passwords in one second, and each passcode is equally likely. Note that we are trying passwords sequentially, not randomly, so using the geometric distribution was not appropriate here.
- (c)

$$\sum_{i=1}^{\infty} \left(P(X=i) \left(\sum_{j=0}^{i-1} 2^j \right) \right) = \sum_{i=1}^{\infty} \left(1 - \frac{1}{1000} \right)^{i-1} \left(\frac{1}{1000} \right) (2^i - 1)$$

3. (a)

$$\binom{2050}{2} = \sum_{n=0}^{2049} n = \sum_{n=1}^{2050} 2050 - n$$

Common mistakes: $2050 \cdot 2049$ (order shouldn't matter), off by 1 on summation bounds, product instead of summation.

(b)

$$\binom{50}{2} = \sum_{n=0}^{49} n = \sum_{n=1}^{50} 50 - n$$

Common mistakes: $50 \cdot 49$ (order shouldn't matter), off by 1 on summation bounds, product instead of summation.

(c)

$$\binom{2050}{2}/5 - \binom{50}{2} + 1 = \binom{2050}{2}/5 - \binom{50}{2}$$

We gave credit for correct setup and technique, even if you used incorrect values found in (a) and (b).

(d)

$$\sum_{i=d}^n \binom{n}{i} \left(\frac{1}{5}\right)^i \left(\frac{4}{5}\right)^{n-i} = 1 - \sum_{i=0}^d \binom{n}{i} \left(\frac{1}{5}\right)^i \left(\frac{4}{5}\right)^{n-i}$$

where $n = \binom{2050}{2} - \binom{50}{2}$. We accepted approximations with a normal distribution as long as it was mentioned that it was an approximation, and a justification was given for why a normal approximation is appropriate. We also accepted bounds beginning or ending at $d + 1$ instead of d . Common mistakes: using Poisson or negative binomial; incorrect bounds; incorrect binomial form; finding the probability that a song receives exactly $1/5$ of votes instead of $> 1/5$.

(e) Let S_2 be the event that Shazam predicts X_2 , the Andy Williams song. $P(S_2 | X_1) = P(\text{Shazam is incorrect}) = 1 - q$, and $P(S_2 | X_2) = P(\text{Shazam is correct}) = q$. Using Bayes' theorem:

$$\begin{aligned} P(X_1 | S_2) &= \frac{P(S_2 | X_1)P(X_1)}{P(S_2 | X_1)P(X_1) + P(S_2 | X_2)P(X_2)} \\ &= \frac{(1-q)0.8}{(1-q)0.8 + q \cdot 0.2} \\ P(X_2 | S_2) &= \frac{P(S_2 | X_2)P(X_2)}{P(S_2 | X_1)P(X_1) + P(S_2 | X_2)P(X_2)} \\ &= \frac{q \cdot 0.2}{(1-q)0.8 + q \cdot 0.2} \end{aligned}$$

Common mistake: trying to find $P(X_1 | \text{Shazam is correct})$. We don't know that the prediction is correct, just that it predicted the Andy Williams song.

4. (a) For $X \sim N(0, 2)$,

$$\begin{aligned} P(X \geq 1.2) &= P\left(\frac{X-0}{\sqrt{2}} \geq \frac{1.2-0}{\sqrt{2}}\right) \\ &= P\left(Z \geq \frac{1.2}{\sqrt{2}}\right) \\ &= 1 - P\left(Z \leq \frac{1.2}{\sqrt{2}}\right) \\ &= 1 - P(Z \leq 0.85) \\ &= 0.1977 \end{aligned}$$

(b) For $X \sim \text{Uni}(0, 10)$ and $Y \sim \text{Uni}(0, 10)$,

$$f_X(x) = \begin{cases} \frac{1}{10} & 0 \leq x \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{10} & 0 \leq y \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

Since X and Y are independent,

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y) = \begin{cases} \frac{1}{100} & 0 \leq x, y \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

- (c) Let R be the event that the user is a robot, C be the position that the user clicked, and D being the position's distance to the center.

$$\begin{aligned} P(C = (3, 3)|R) &= \frac{P(C = (3, 3)|R)P(R)}{P(C = (3, 3)|R)P(R) + P(C = (3, 3)|R^C)P(R^C)} \\ &= \lim_{\varepsilon \rightarrow 0} \frac{\varepsilon f_{D|R}(2)P(R)}{\varepsilon f_{D|R}(2)P(R) + \varepsilon f_{C|R^C}((3, 3))P(R^C)} \\ &= \lim_{\varepsilon \rightarrow 0} \frac{f_{D|R}(2)P(R)}{f_{D|R}(2)P(R) + f_{C|R^C}((3, 3))P(R^C)} \\ &= \frac{f_{C|R}(2)0.2}{f_{D|R}(2)0.2 + f_{C|R^C}((3, 3))0.8} \\ &= \frac{\frac{1}{\sqrt{4\pi}} e^{\frac{(2-0)^2}{2 \cdot 2}} 0.2}{\frac{1}{\sqrt{4\pi}} e^{\frac{(2-0)^2}{2 \cdot 2}} 0.2 + \frac{1}{100} 0.8} \\ &= \frac{\frac{1}{\sqrt{4\pi}} e^{-1} 0.2}{\frac{1}{\sqrt{4\pi}} e^{-1} 0.2 + \frac{1}{100} 0.8} \end{aligned}$$

Since $D = 2$ is not exactly equivalent to $C = (3, 3)$, we also gave credit if you divided $f_{D|R}(2)$ by 4π , the circumference of the circle with radius 2!

5. (a) Standard deviation = $\sqrt{\text{Var}(X)} = \frac{1}{\lambda}$
Mean = $E[X] = \frac{1}{\lambda}$
Using the CDF of an exponential $F_X(x) = 1 - e^{-\lambda x}$, we get $P(0 < x < \frac{2}{\lambda}) = F_X(\frac{2}{\lambda}) = 1 - e^{-2}$.
- (b) Let M be the maximum of X_1 and X_2 . The CDF of M , $F_M(k)$, is the probability that both X_1 and X_2 are at most k , so we need:

$$\begin{aligned} P(M \leq k) &= P(X_1 \leq k \cap X_2 \leq k) \\ &= P(X_1 \leq k) \cdot P(X_2 \leq k) && \text{by independence} \\ &= (1 - e^{-k\lambda_1})(1 - e^{-k\lambda_2}) \end{aligned}$$

6. (a) $P(\text{no drop in any given minute}) = 1 - p$. The number of requests in each minute, and therefore the probability of dropping a request in each minute, are independent random variables, so

$$\begin{aligned} P(\text{no drops in whole hour}) &= (1 - p)^{60} > 0.99 \\ p &< 1 - (0.99)^{\frac{1}{60}} \\ &\left[\text{or } 1 - \sqrt[60]{0.99}, \text{ or } 1 - e^{\left(\frac{1}{60} \ln 0.99\right)} \right] \end{aligned}$$

This is equal to $P(X = 0)$ for $X \sim \text{Bin}(60, p)$. (Common mistake: using 15 million as the number of trials for this binomial.) Since $n = 60$ is large enough and p is small, we gave almost full credit for a correct Poisson approximation ($p < -\frac{1}{60} \ln 0.99$). However, note that approximation is unnecessary here and doesn't really simplify the answer.

- (b) Let Y be a random variable representing the number of requests received in a particular minute. $Y \sim \text{Poi}(\lambda)$, with $\lambda = 15,000,000$. Each server can handle $C = 10,000$ requests per minute, so the probability of dropping a request in any given minute is

$$\begin{aligned} p &= P(Y > CK) = 1 - P(Y \leq CK) \\ &= 1 - \sum_{i=0}^{CK} p_Y(i) \\ &= 1 - \sum_{i=0}^{CK} e^{-\lambda} \frac{\lambda^i}{i!} \end{aligned}$$

From part (a) we have $p < 1 - (0.99)^{\frac{1}{60}}$, so

$$\begin{aligned} p &= 1 - \sum_{i=0}^{CK} e^{-\lambda} \frac{\lambda^i}{i!} < 1 - (0.99)^{\frac{1}{60}} \\ \sum_{i=0}^{10,000K} e^{-15,000,000} \frac{15,000,000^i}{i!} &> (0.99)^{\frac{1}{60}} \end{aligned}$$

(There isn't a closed-form solution for K in terms of elementary operations; we gave full credit for getting to this inequality.)

Part (d) explores the use of the normal approximation, which is applicable for this value of λ , but we asked for an exact solution in part (a), so we took off a bit more for using an approximation here.

- (c) We start by approximating $X \sim P(1000)$ with $Y \sim N(1000, 1000)$:

$$\begin{aligned} p(990 < X < 1000) &= P(990.5 < Y < 999.5) && \text{continuity correction} \\ &= P(Y < 999.5) - P(Y < 990.5) \\ &= P\left(\frac{Y - 1000}{\sqrt{1000}} < \frac{999.5 - 1000}{\sqrt{1000}}\right) - \\ &\quad P\left(\frac{Y - 1000}{\sqrt{1000}} < \frac{990.5 - 1000}{\sqrt{1000}}\right) \\ &= P\left(Z < \frac{-0.5}{\sqrt{1000}}\right) - P\left(Z < \frac{-9.5}{\sqrt{1000}}\right) \\ &= \Phi\left(\frac{-0.5}{\sqrt{1000}}\right) - \Phi\left(\frac{-9.5}{\sqrt{1000}}\right) \\ &= 1 - \Phi\left(\frac{0.5}{\sqrt{1000}}\right) - \left(1 - \Phi\left(\frac{9.5}{\sqrt{1000}}\right)\right) \\ &= \Phi\left(\frac{9.5}{\sqrt{1000}}\right) - \Phi\left(\frac{0.5}{\sqrt{1000}}\right) \end{aligned}$$

- (d) Let X be a random variable which represents the number of requests in a minute. Since we receive on average 15 million requests per minute, $X \sim \text{Poi}(\lambda = 15 \text{ million})$.

Following from part (c), since the value of λ for X is larger than 1,000, we can approximate X using a normal that matches its mean and variance. Since both the mean and variance of a Poisson are equal to λ , $X \approx Y$ where $Y \sim N(\mu = 15 \text{ million}, \sigma^2 = 15 \text{ million})$.

We want to choose number of servers K such that the probability that we get more than $10,000 \cdot K$ requests in one minute is equal to p from part (a):

$$\begin{aligned}
p &= P(X \leq 10\,000K) \\
&\approx P(Y < 10\,000K + 0.5) && \text{continuity correction} \\
&= P\left(Z < \frac{10\,000K + 0.5 - \lambda}{\sqrt{\lambda}}\right) && \text{transform to standard normal} \\
&= \Phi\left(\frac{10\,000K + 0.5 - \lambda}{\sqrt{\lambda}}\right) && \text{since } \Phi \text{ is the CDF of } Z \\
\Phi^{-1}(p) &= \frac{10\,000K + 0.5 - \lambda}{\sqrt{\lambda}} && \text{using the probit function} \\
K &= \frac{\Phi^{-1}(p)\sqrt{\lambda} + \lambda - 0.5}{10\,000}
\end{aligned}$$

The expression above would get full credit. For curiosity, let's find out the numeric answer:

$$\begin{aligned}
K &= \frac{\Phi^{-1}(p)\sqrt{\lambda} + \lambda - 0.5}{10\,000} \\
K &= \frac{\Phi^{-1}(\sqrt[60]{0.99})\sqrt{15\,000\,000} + 15\,000\,000 - 0.5}{10\,000} \\
K &= \frac{3.5866 \cdot 3872.98 + 15\,000\,000 - 0.5}{10\,000} \\
K &= \lceil 1501.4 \rceil = 1502
\end{aligned}$$

Interestingly, with 1500 servers, you have a 0.5 chance of dropping a request. With just two more servers the probability of not-dropping a request increases to 0.9998 (which is the numeric value for p).